Math 55 worksheet Monday 03/30/2026

1. Count the following collections:
   - The number of integers $k \in \{1, 2, ..., 100\}$ so that $2 \mid k$ or $5 \mid k$.
   - The number of three-letter strings (e.g. ABC, XYZ).
   - The number of ways of splitting a group of 10 people into two groups of five.

2. Count strings of three letters followed by three digits with no repeated characters (e.g. $ABC123$)?

3. How many 5 letter strings are there with at least one vowel?

4. Show that the number a_n of subsets of $\{1, 2, ..., n\}$ that do not contain consecutive integers equals the Fibonacci number f_{n+2}. Recall that f_n is defined via $f_n = f_{n-1} + f_{n-2}$ with $f_1 = f_2 = 1$.

5. For fixed $1 \leq k \leq n$, how many distinct cycles of length k are there in the complete graph K_n?

6. Define a simple walk of length $2n$ as a sequence of integers $(x_0, x_1, x_2, ..., x_{2n})$ so that $(x_{i+1} - x_i) \in \{-1, 1\}$ for all i. For example, $(0, 1, 2, 1, 0)$ is a simple walk of length four.

   (i) How many simple walks are there of length $2n$ that end at zero ($x_{2n} = 0$)?

   (ii) How many simple walks are there of length $2n$ that end at zero and are always nonnegative? Hint[1]

[1] count the number of walks not satisfying this property. Try to relate this count to walks from 0 to -2 in $2n$ steps